Name the force and its source

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Identify forces on a chosen object and name each interaction partner.

SUCCESS CRITERIA - I can...

1. Name the chosen object and each force source.
2. Distinguish mass from weight with units.
3. Include only forces acting on the chosen object.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “interaction” and “weight” to explain your answer. Sentence starters are optional.

interaction: two objects affecting one another

weight: the gravitational force on an object

normal force: a support force perpendicular to a contact surface

01 Warm-up: bring back an earlier idea

A cart travels *west* at **12 m/s** and *west* at **4 m/s 3 s** later. Is it speeding up? **Find acceleration size** and direction.

Word help: acceleration = rate of change of velocity

Sentence starter: I found ... [unit]. This value means ... in this situation.

Marzano check - circle: ① ② ③

02 Worked example: explain the prediction

Reference: $F_g = mg$.
F_g is weight; m is mass. Use g = **10 N/kg** for CPA. **Name the source of every** force.

Worked reasoning: A book rests on a table. Earth pulls the book down; the table pushes the book up. Each force label names both the force and the object producing it.

Your turn: 1. **Draw the book's force diagram**.
2. **Name the force** on Earth paired with the book's weight.

Sentence starter: In my work, I used ... to show ... . I labeled it ... .



Add and label force arrows.

Book resting on a level table; draw forces on the book only.

Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

03 Which object gets the force?

A book rests on a level table. **Draw its two vertical forces** and **name who exerts each**.

Sentence starter: In my work, I used … to show … . I labeled it … .

Add and label force arrows.

The dot represents the object. Chosen object: book

Marzano check - circle: 1 2 3

04 Mass and weight

A **2 kg** bag hangs at rest from a vertical rope.
1. Use g = **10 N/kg**.
2. **Draw the forces** on the bag, then **find its weight** and the rope tension.
3. **Label each force** with its source.

Word help: at rest = not moving in the chosen reference frame

Sentence starter: I found … [unit]. This value means … in this situation.

Add and label force arrows.

Chosen object: hanging bag.

Marzano check - circle: 1 2 3

05 After the throw

A ball is moving upward after it leaves your hand. Ignore air resistance.
1. **Draw its force diagram** and **name the source of every** force.
2. **Explain why upward motion** does not require an upward force.

Sentence starter: I think … . I can point to …, which shows …, so … .

Add and label force arrows.

Ball after release; velocity is upward.

Marzano check - circle: 1 2 3

Name the force and its source

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Continue down each column.

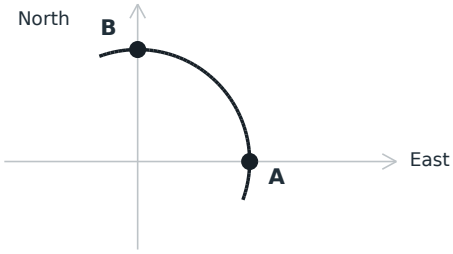
Sentence starters are optional. You may plan in any language.

06 Review: connect an earlier idea

A rover travels straight *east* at a steady **5 m/s**. A second rover turns at the same steady speed. Which can have zero net force? Explain using velocity.
Use the curve for the turning case; **show how velocity changes**.

Word help: net force = the vector sum of all forces on one object

Sentence starter: I think … . I can point to …, which shows …, so … .



Path sketch: runner moves from A toward B.

Marzano check - circle: 1 2 3

07 Exit check

A **3 kg** lamp hangs still from one vertical cord.
1. Use g=**10 N/kg**.
2. **Name both forces** with source, magnitude and direction.

Word help: magnitude = size of a quantity without its direction

Sentence starter: I notice … . This supports … because … .

Add and label force arrows.

Hanging object at rest: label sources and magnitudes.

Marzano check - circle: 1 2 3

08 STRETCH: test the prediction

A hand presses down on the resting book. **Add the new force** and **explain how the table's support** must change.

Sentence starter: I think … . I can point to …, which shows …, so … .

Add and label force arrows.

Book pressed down by a hand: draw forces on the book.

Marzano check - circle: 1 2 3